

An Analysis of AlphaFold

AlphaFold

AlphaFold -- Part 1

CASP13 In December 2018, DeepMind's AlphaFold placed first in the overall rankings of the 13th Critical Assessment of Techniques for Protein Structure Prediction (CASP). The program was particularly successfully predicting the most accurate structure for targets rated as the most difficult by the competition organisers, where no existing template structures were available from proteins with a partially similar sequence. AlphaFold gave the best prediction for 25 out of 43 protein targets in this class, achieving a median score of 58.9 on the CASP's global distance test (GDT) score, ahead of 52.5 and 52.4 by the two next best-placed teams, who were also using deep learning to estimate contact distances. Overall, across all targets, AlphaFold 1 achieved a GDT score of 68.5. In January 2020, implementations and illustrative code of AlphaFold 1 was released open-source on GitHub. but, as stated in the "Read Me" file on that website: "This code can't be used to predict structure of an arbitrary protein sequence. It can be used to predict structure only on the CASP13 dataset (links below). The feature generation code is tightly coupled to our internal infrastructure as well as external tools, hence we are unable to open-source it." Therefore, in essence, the code deposited is not suitable for general use but only for the CASP13 proteins. The company has not announced plans to make their code publicly available as of 5 March 2021.

AlphaFold -- Part 2

AlphaFold 1 (2018) was built on work developed by various teams in the 2010s, work that looked at the large databases of related protein sequences now available from many different organisms (most without known 3D structures), to try to find changes at different residues (peptides) that appeared to be correlated, even though the residues were not consecutive in the main chain. Such correlations suggest that the residues may be close to each other physically, even though not close in the sequence, allowing a contact map to be estimated. Building on recent work prior to 2018, AlphaFold 1 extended this by estimating a probability distribution for the distances between residues, effectively transforming the contact map into a distance map. It also used more advanced learning methods than previously to develop the inference. The code was not made publicly available, except to run on sequences of proteins in the 2018 CASP competition.

AlphaFold -- Part 3

Algorithm DeepMind is known to have trained the program on over 170,000 protein structures from the Protein Data Bank, a public repository of protein sequences and structures. The

program uses a form of attention network, a deep learning technique that focuses on having the AI identify parts of a larger problem, then piece it together to obtain the overall solution. The overall training was conducted on processing power between 100 and 200 GPUs.

AlphaFold -- Part 4

Further reading Carlos Outeiral, CASP14: what Google DeepMind's AlphaFold 2 really achieved, and what it means for protein folding, biology and bioinformatics, Oxford Protein Informatics Group. (3 December) Mohammed AlQuraishi, AlphaFold2 @ CASP14: "It feels like one's child has left home." (blog), 8 December 2020 Mohammed AlQuraishi, The AlphaFold2 Method Paper: A Fount of Good Ideas (blog), 25 July 2021

AlphaFold -- Part 5

Proteins consist of chains of amino acids which spontaneously fold to form the three dimensional (3-D) structures of the proteins. The 3-D structure is crucial to understanding the biological function of the protein. Protein structures can be determined experimentally through techniques such as X-ray crystallography, cryo-electron microscopy and nuclear magnetic resonance (NMR), which are all expensive and time-consuming. Such efforts, using the experimental methods, have identified the structures of about 170,000 proteins over the last 60 years, while there are over 200 million known proteins across all life forms. Over the years, researchers have applied numerous computational methods to predict the 3D structures of proteins from their amino acid sequences, accuracy of such methods in best possible scenario is close to experimental techniques (NMR) by the use of homology modeling based on molecular evolution. CASP, which was launched in 1994 to challenge the scientific community to produce their best protein structure predictions, found that GDT scores of only about 40 out of 100 can be achieved for the most difficult proteins by 2016. AlphaFold started competing in the 2018 CASP using an artificial intelligence (AI) deep learning technique.

AlphaFold -- Part 6

Derived databases AlphaFill adds cofactors to AlphaFold models where appropriate. This is achieved by searching the Protein Data Bank for similar structures and transplanting cofactors to analogous positions. It is also linked to by UniProt. TmAlphaFold docks AlphaFold models to biological membranes, similar to what OPM does for PDB structures. AFTM uses AlphaFold models to identify transmembrane regions in human proteins, similar to what PDBTM does for PDB

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structures. ChannelsDB 2.0 uses PDB or AlphaFill models to calculate the pathway a molecule may take to reach an enzyme's active site or to reach another side of a transmembrane pore. The AFDB is not updated with UniProt sequences changes. AlphaSync keeps the AFDB in sync with UniProt entry changes, generating updated structures, residue-level features and contacts. It tries to use an AFDB entry for the exact updated sequence when available and run AlphaFold 2 independently otherwise. It fills in AFDB's blank for large (> 2,700 aa) proteins and proteins with special FASTA characters such as B, Z, U or X. The Encyclopedia of Domains (TED) applies the domain-recognition method from CATH database to 188 million unique structures from the AFDB, identifying nearly 365 million domains, which is 100 million more than what sequence-based methods could identify. The Evolutionary Classification of Protein Domains database (ECOD) assigns ECOD classifications to all SwissProt proteins in AFDB.

AlphaFold -- Part 7

Clones and derivatives A number of AlphaFold clones have also been published, mostly with permissive license terms. Clones for AlphaFold3 include ByteDance's Protenix (Apache 2.0 License), AlQuraishi Laboratory's OpenFold-3 (MIT license), and Boltz-1/2 (MIT license). There are also clones for older versions, though they became less relevant with the open-source release of AlphaFold 1 and 2 source codes. Still relevant are models, both open- and closed-source, that include modifications to the AlphaFold architecture. For AlphaFold 2, a notable example is ESMFold from Meta, which replaces the multiple sequence alignment with the latent space of a protein language model. Open-source tools that complement AlphaFold have also been made. One well-cited example is ColabFold, which uses MMseqs2 instead of HHblits to speed up the sequence search, allowing the AlphaFold pipelines to run quickly on Google Colab.